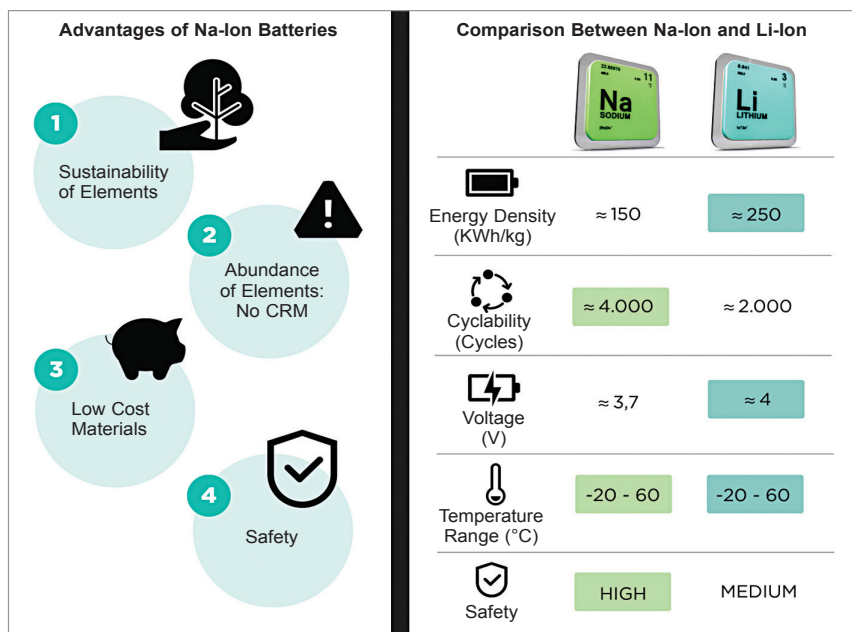


Anode materials. Insertion anode graphite, commonly used in LIBs with carbonate-based electrolytes, does not work in SIBs due to the larger size of the Na^+ ions. Hard carbon is the only insertion anode material being used commercially. Other insertion anode materials, such as sodium titanate and sodium titanium phosphate, are also being studied for future use. The limited specific capacity in insertion materials poses serious concerns. Alloying anodes, such as Sn and Sb, can provide higher specific capacity. However, the severe volume change during the alloying reaction due to the large size of Na^+ ultimately leads to cell failure. Recently, research has shifted from alloying and insertion materials to conversion materials, such as metal oxides, metal sulphides, and metal selenides. Conversion materials facilitate fast kinetics with a higher potential (lower overall voltage) compared to alloying and insertion anodes and provide the highest power density compared to alloying and insertion materials. Insertion materials are the most energy dense, followed by alloying and conversion materials. The fast reaction in conversion materials favours power density, while the lower potential (higher overall voltage) of insertion anode materials favours energy density. It is essential to optimise between power density and energy density in the electrode chemistries. Research on organic anode materials, another class of anodes, is still in the nascent stage.

Cathode materials. Cathode materials play a critical role in the performance of SIBs. The four main types of insertion cathodes, categorised by structure, are sodium transition metal oxide, polyanionic, Prussian blue analogue, and organic-based cathodes. Amongst the insertion materials, sodium iron hexacyanoferrate has been found to be the most promis-



Advantages of SIBs and comparison with LIBs (Credit: <https://cicenergigune.com>)

ing in terms of energy density, and phosphate-based sodium vanadium in terms of fast charging. Sodium analogue cathodes, specifically metal oxides and polyanionic materials, perform adequately with energy densities above 100Wh/kg and voltages over 3.0V. Many start-ups are working on Prussian blue analogues to avoid the use of cobalt in the cathodes. The nanostructuring of the materials can alleviate most of the limitations, enhancing performance in terms of capacity and rate capability. However, there are certain concerns such as structural instability, limited ionic movement, and low electrical conductivity that need to be addressed. The formation of materials with a carbon framework that has nanostructuring helps in this regard.

Electrolytes. The choice of electrolytes is important, and both aqueous and non-aqueous electrolytes are used in SIBs. Aqueous electrolytes provide limited electrochemical stability, lower voltages, and energy densities. Non-aqueous carbonate ester polar aprotic solvents such as ethylene carbonate, dimethyl

carbonate, diethyl carbonate, and propylene carbonate can extend the voltage range. Sodium hexafluorophosphate is added to a mixture of these solvents to stabilise the anode and cathode during cycling. Mixed organic carbonate solvents provide optimal discharge and charge over a wide temperature range, thermal stability, high-rate capability, long cycle life, and shelf life.

How SIBs compare with LIBs?

SIBs are a promising alternative to LIBs, the most widely used rechargeable batteries. The build and performance of both LIBs and SIBs are almost similar. Both have similar power delivery characteristics and use liquid electrolyte to store and transfer electrical energy with different types of ions. The lower dissolving energy for sodium ions, i.e., 157kJ/mol compared to 218kJ/mol for lithium ions, enhances the kinetics in SIBs, allowing for faster charging rates and a longer cycle life. SIBs have slightly lower energy density but better safety characteristics. The biggest advantage of SIBs is the high natural abundance